2020: GPT3

Causal Language model

Prompt-based Learning

Tanmoy Chakraborty
Associate Professor, IIT Delhi
https://tanmoychak.com/

Many slides from Mohit Iyyer, Graham Neubig

Recommended Reading

Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing

Pengfei Liu
Carnegie Mellon University
pliu3@cs.cmu.edu

Weizhe Yuan
Carnegie Mellon University
weizhey@cs.cmu.edu

Jinlan Fu
National University of Singapore
jinlanjonna@gmail.com

Zhengbao Jiang
Carnegie Mellon University
zhengbaj@cs.cmu.edu

Hiroaki Hayashi
Carnegie Mellon University
hiroakih@cs.cmu.edu

Graham Neubig
Carnegie Mellon University
gneubig@cs.cmu.edu

The Language Model "Scaling Wars"!

RNN

ELMo: 93M params, 2-layer biLSTM

2017

BERT-base: 110M params, 12-layer Transformer

BERT-large: 340M params, 24-layer Transformer

Q
K
V

$w \leftarrow w + \eta \frac{\delta L}{\delta w}$

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or "GPT-3"	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

The Language Model "Scaling Wars"!

ELMo: 93M params, 2-layer biLSTM

BERT-base: 110M params, 12-layer Transformer

BERT-large: 340M params, 24-layer Transformer

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or "GPT-3"	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

The Language Model "Scaling Wars"!

ELMo: 1B training tokens

BERT: 3.3B training tokens

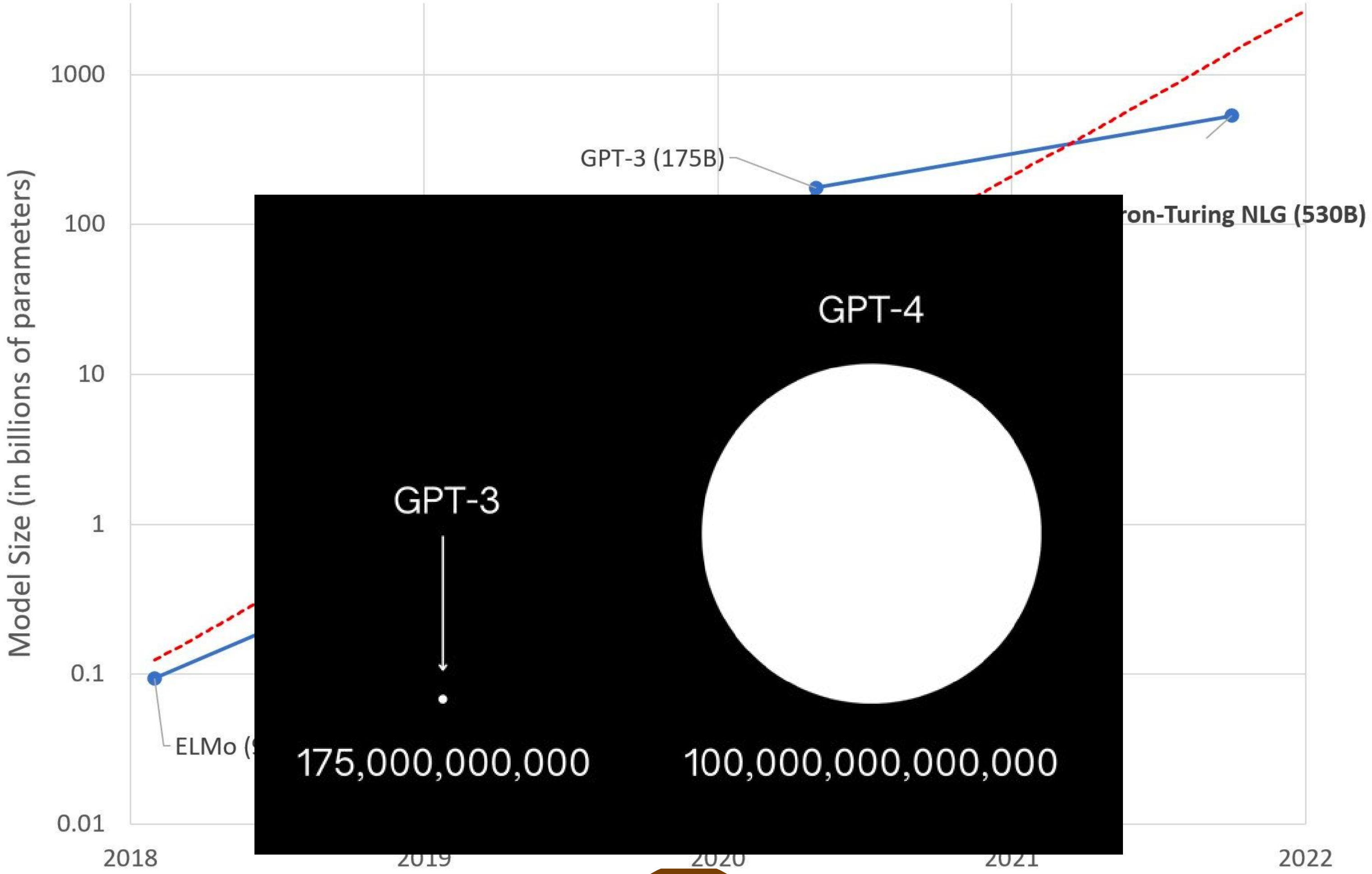
RoBERTa: ~30B training tokens

T5

C4

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

Colossal Models



So… What Does All of This Scaling Buy Us?

(i) Few-shot

LLMs

(ii)

100 examples

10K ex..

Pre-trained Model

Zero-Shot learning

2-shot learning

I want to perform sentiment analysis.
Here are some examples:
The ... => Positive
The ... => Neg.
Tell me the sent of the f...
The shop was horrible => GPT-3

GPT-3

Language Models are Few-Shot Learners

Tom B. Brown* Benjamin Mann* Nick Ryder* Melanie Subbiah*

Jared Kaplan† Prafulla Dhariwal Arvind Neelakantan Pranav Shyam Girish Sastry

Amanda Askell Sandhini Agarwal Ariel Herbert-Voss Gretchen Krueger Tom Henighan

Rewon Child Aditya Ramesh Daniel M. Ziegler Jeffrey Wu Clemens Winter

Christopher Hesse Mark Chen Eric Sigler Mateusz Litwin Scott Gray

Benjamin Chess Jack Clark Christopher Berner

Sam McCandlish Alec Radford Ilya Sutskever Dario Amodei

1 - Only Pre-trained model without example (0-shot)

2 - Only Pre-trained model with few-shot

3 - Fine-tuned model.

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.

Downstream training data

Downstream test data

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
Translate English to French:    ← task description
cheese =>                        ← prompt
```

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
Translate English to French:     ← task description
cheese =>                        ← prompt
```

=> :

No fine-tuning!!! Literally just take a pretrained LM and give it the following prefix:

"**Translate English to French: cheese =>**"

We will see how LLMs are very 'sensitive' to such prompt formatting, and how we can measure this sensitivity!

Why "=>"? What is the optimal prompt?

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1  Translate English to French:      ← task description
2  sea otter => loutre de mer         ← example
3  cheese =>                          ← prompt
```

No fine-tuning!!! Literally just take a pretrained LM and give it the following prefix:

"Translate English to French: sea otter => loutre de mer, cheese =>"

No fine-tuning!!! Literally just take a pretrained LM and give it the following prefix:

Many such examples fed into the prefix in this way

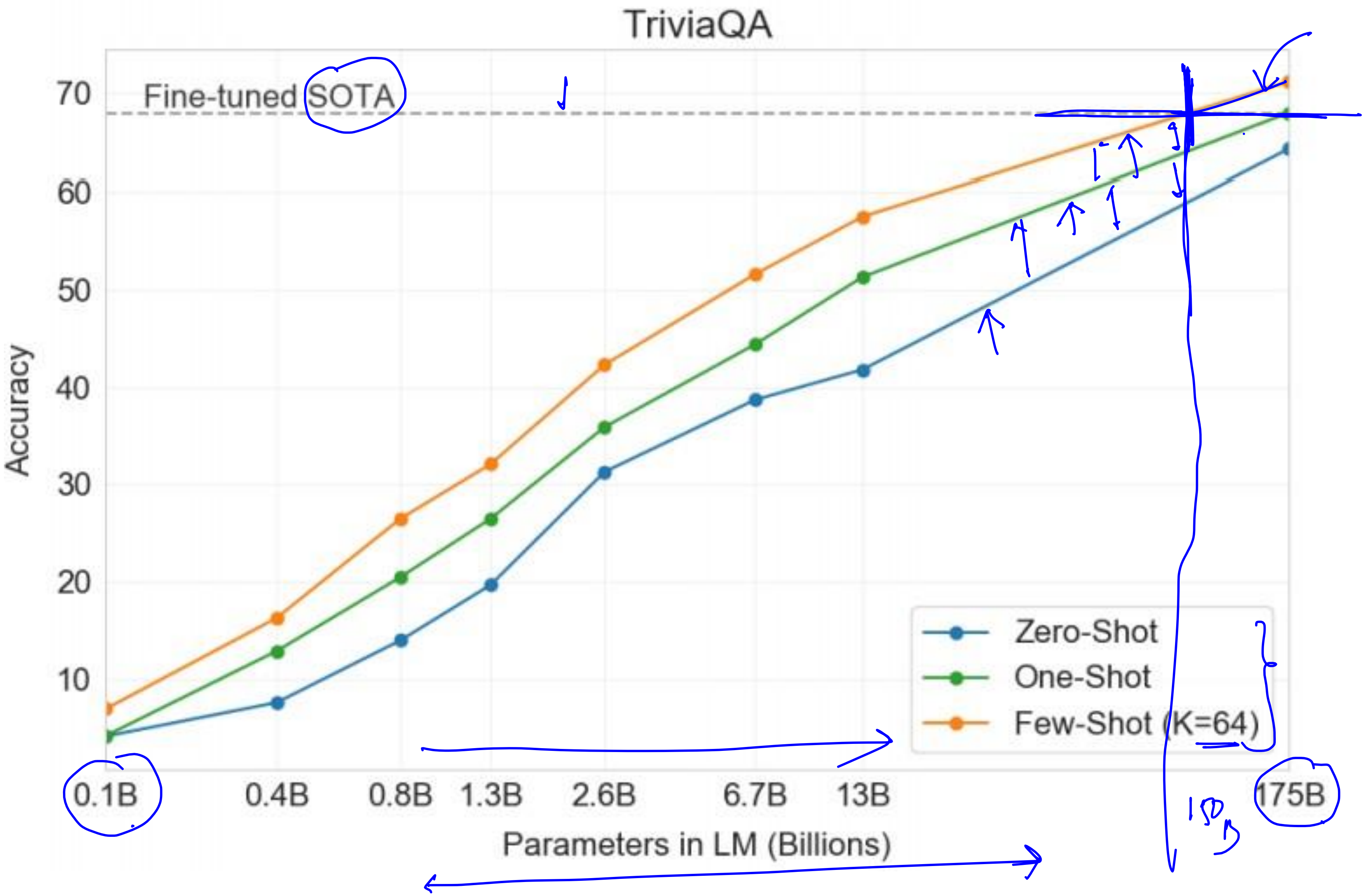
"Translate English to French: sea otter => loutre de mer, peppermint => … (few more examples) , cheese => "

How Does This New Paradigm Compare to "Pretrain + Finetune"?

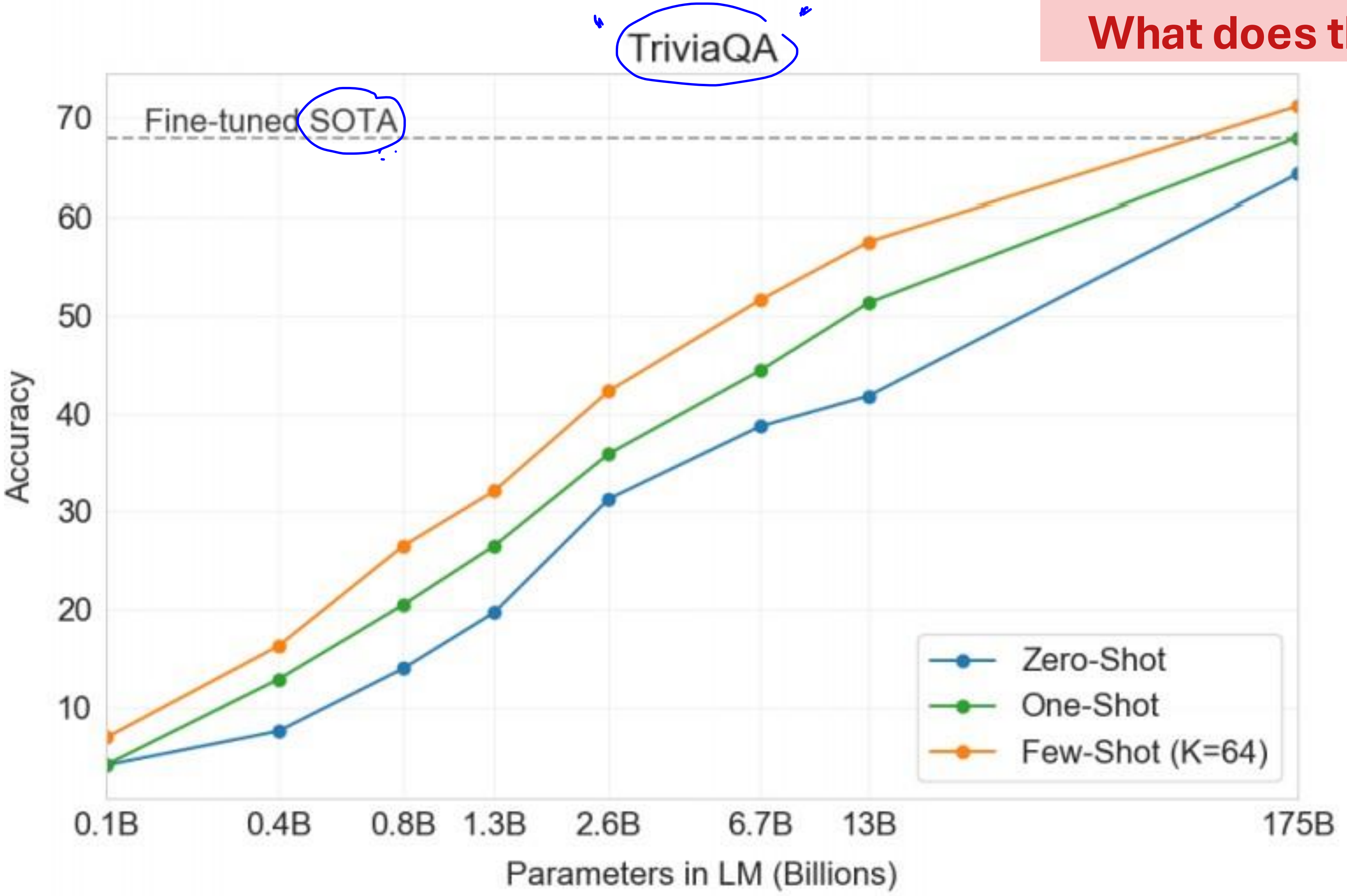
TriviaQA

Hotpot QA

Question
Miami Beach in Florida borders which ocean?
What was the occupation of Lovely Rita according to the song by the Beatles
Who was Poopdeck Pappys most famous son?
The Nazi regime was Germany's Third Reich; which was the first Reich?
At which English racecourse did two horses collapse and die in the parade ring due to electrocution, in February 2011?
Which type of hat takes its name from an 1894 novel by George Du Maurier where the title character has the surname O'Ferrall ?
What was the Elephant Man's real name?


TriviaQA
Fine-tuned SOTA
70
60
50
40
30
20
10
Accuracy
Zero-Shot
One-Shot
Few-Shot (K=64)
0.1B
0.4B
0.8B
1.3B
2.6B
6.7B
13B
175B
Parameters in LM (Billions)
150B

What does this mean?

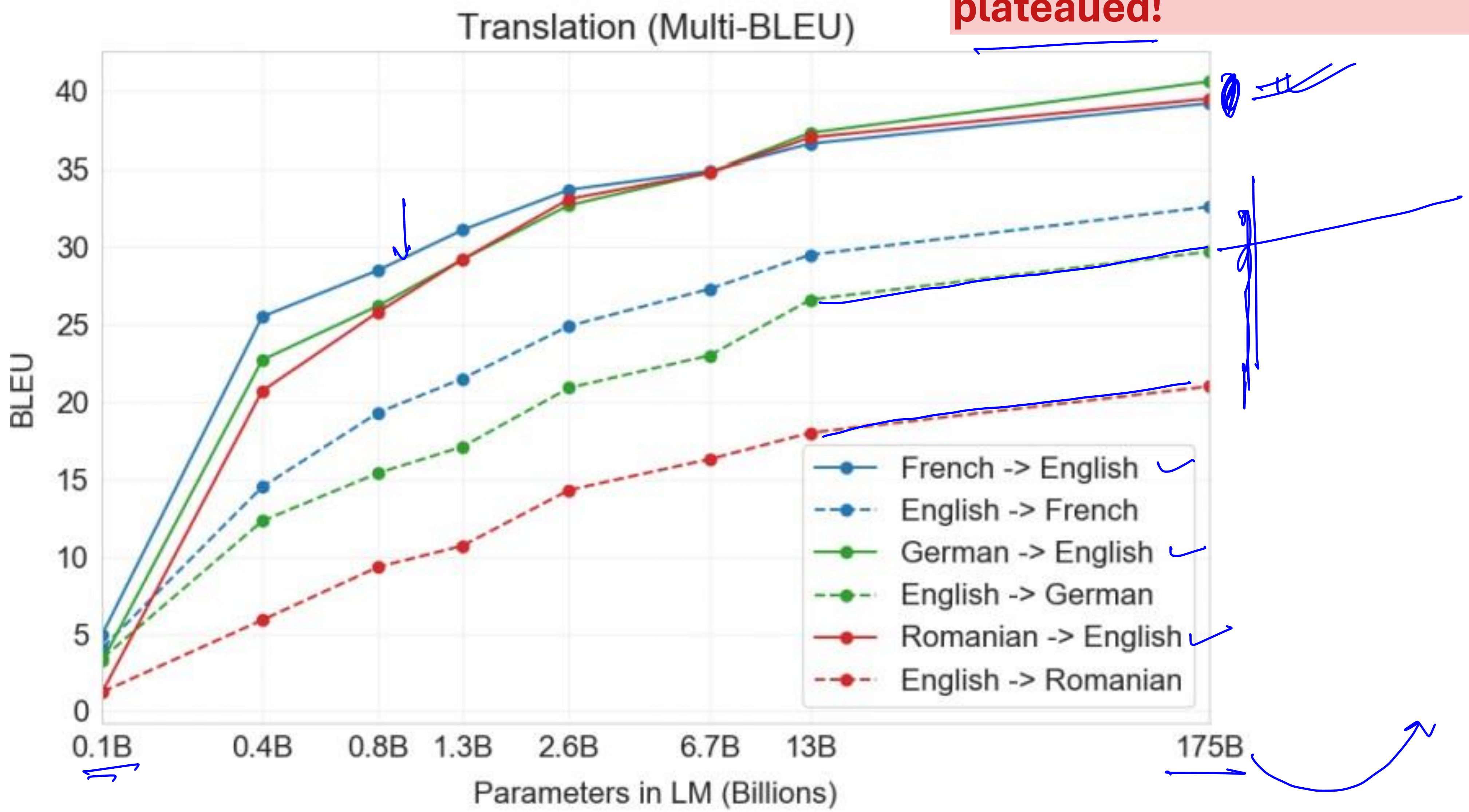


What About Translation?
(7% of GPT3's Training Data is in Languages Other Than English)

BLEU

Setting	En→Fr	Fr→En	En→De	De→En	En→Ro	Ro→En
SOTA (Supervised) *	**45.6**[a]	35.0 [b]	**41.2**[c]	40.2[d]	**38.5**[e]	**39.9**[e]
XLM [LC19]	33.4	33.3	26.4	34.3	33.3	31.8
MASS [STQ+19]	37.5	34.9	28.3	35.2	35.2	33.1
mBART [LGG+20]	-	-	29.8	34.0	35.0	30.5
GPT-3 Zero-Shot	25.2	21.2	24.6	27.2	14.1	19.9
GPT-3 One-Shot	28.3	33.7	26.2	30.4	20.6	38.6
GPT-3 Few-Shot	32.6	39.2	29.7	40.6	21.0	39.5

Improvements haven't plateaued!

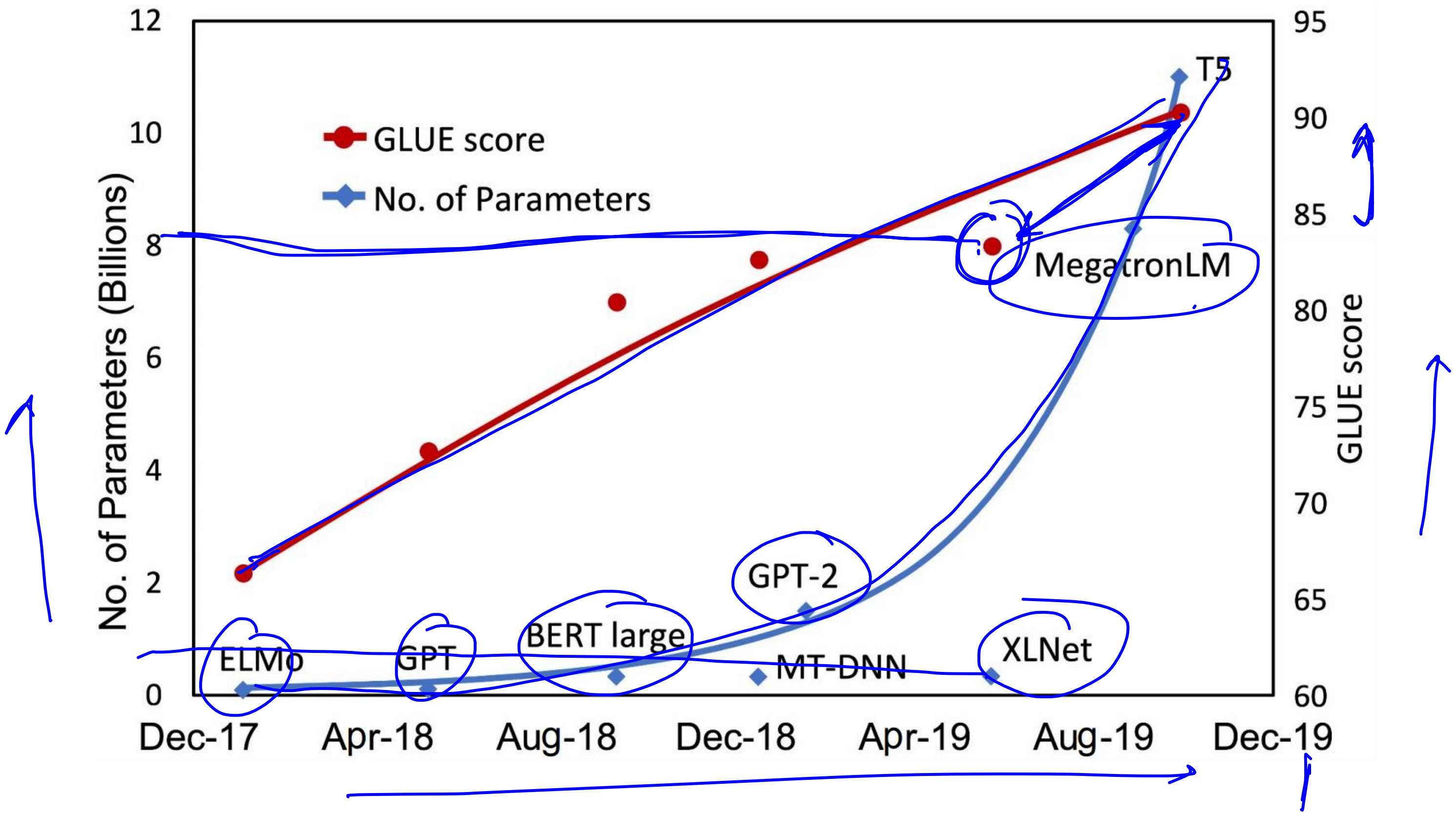


What About Reading Comprehension QA?

Setting	CoQA	DROP	QuAC	SQuADv2	RACE-h	RACE-m
Fine-tuned SOTA	**90.7**[a]	**89.1**[b]	**74.4**[c]	**93.0**[d]	**90.0**[e]	**93.1**[e]
GPT-3 Zero-Shot	81.5	23.6	41.5	59.5	45.5	58.4
GPT-3 One-Shot	84.0	34.3	43.3	65.4	45.9	57.4
GPT-3 Few-Shot	85.0	36.5	44.3	69.8	46.8	58.1

Struggles on "harder" datasets

Scaling up the model size is one of the most important ingredients for achieving the best performance



Ahmet and Abdullah., 2021

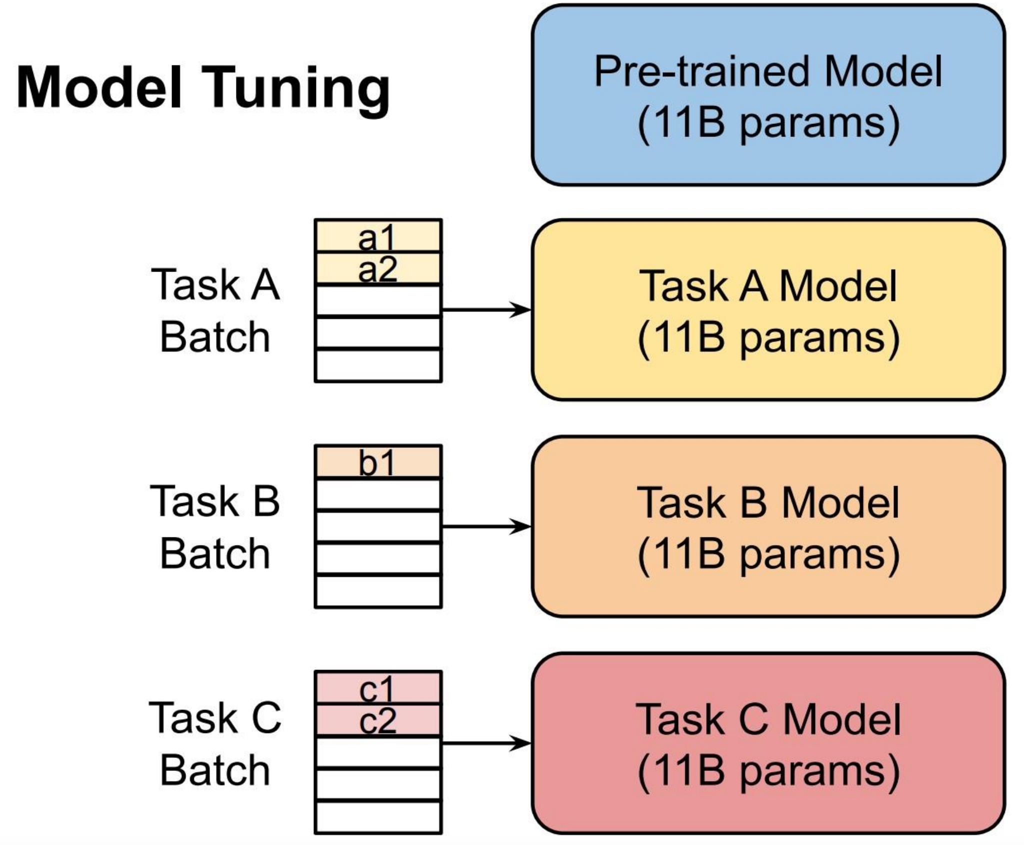
Practical Challenges: Large-Scale Models are Costly to Share and Serve

Prefix tuning

Prompt tuning

~~Hard Prompting~~ →

Soft Prompting



Lester et al., 2021

Language Model Prompting to The Rescue!

GPT-3 ([Brown et al., 2020](#)): **In-context learning**

- **natural language instruction** and/or **a few task demonstrations** ⟶ **output**

  "Translate English to German:" That is good ⟶ Das is gut

- *no* gradient updates or fine-tuning

Sub-optimal and Sensitive Discrete/Hard Prompts

- **Discrete/hard prompts**
  - natural language instructions/task descriptions
- **Problems**
  - requiring domain expertise/understanding of the model's inner workings
  - performance still lags far behind SoTA model tuning results
  - sub-optimal and sensitive
    - prompts that humans consider reasonable is not necessarily effective for language models
    - pre-trained language models are sensitive to the choice of prompts

Sub-optimal and Sensitive Discrete/Hard Prompts

Prompt	P@1
[X] is located in [Y]. *(original)*	31.29
[X] is located in which country or state? [Y].	19.78
[X] is located in which country? [Y].	31.40
[X] is located in which country? In [Y].	51.08

Table 1. Case study on LAMA-TREx P17 with bert-base-cased. A single-word change in prompts could yield a drastic difference.

Liu et al., 2021

Shifting From Discrete/Hard to Continuous/Soft Prompts

Progress in prompt-based learning

- manual prompt design (Brown et al., 2020; Schick and Schutze, 2021a,b)
- mining and paraphrasing based methods to automatically augment the prompt sets (Jiang et al., 2020)
- gradient-based search for improved discrete/hard prompts (Shin et al., 2020)
- automatic prompt generation using a separate generative language model (i.e., T5) (Gao et al., 2020)
- learning continuous/soft prompts (Liu et al., 2021; Li and Liang., 2021; Qin and Eisner., 2021; Lester et al., 2021)

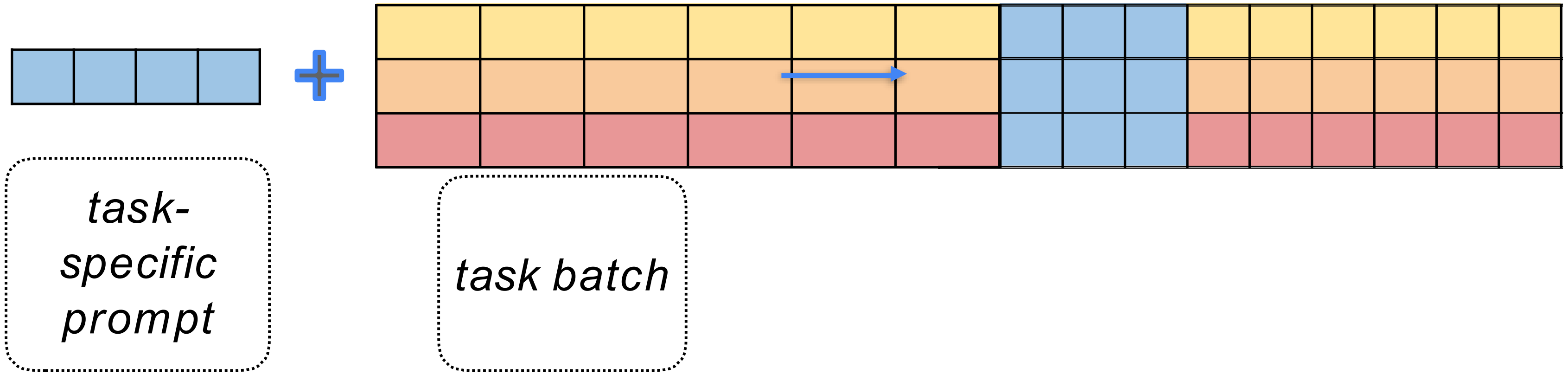
Continuous/soft prompts

- additional learnable parameters injected into the model

Prompt Tuning Idea

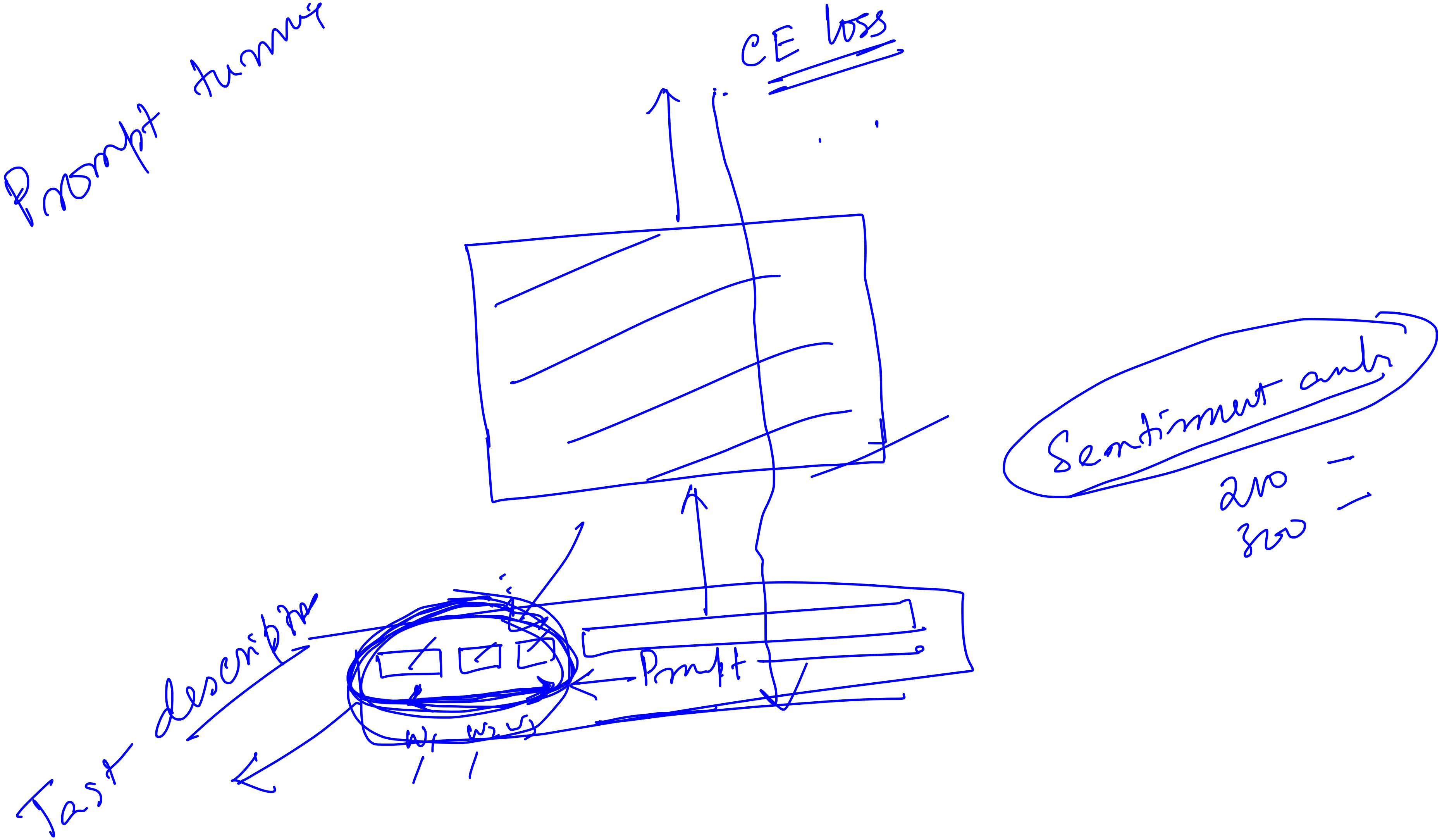
What is a prompt in Prompt Tuning?

A sequence of additional task-specific tunable tokens prepended to the input text

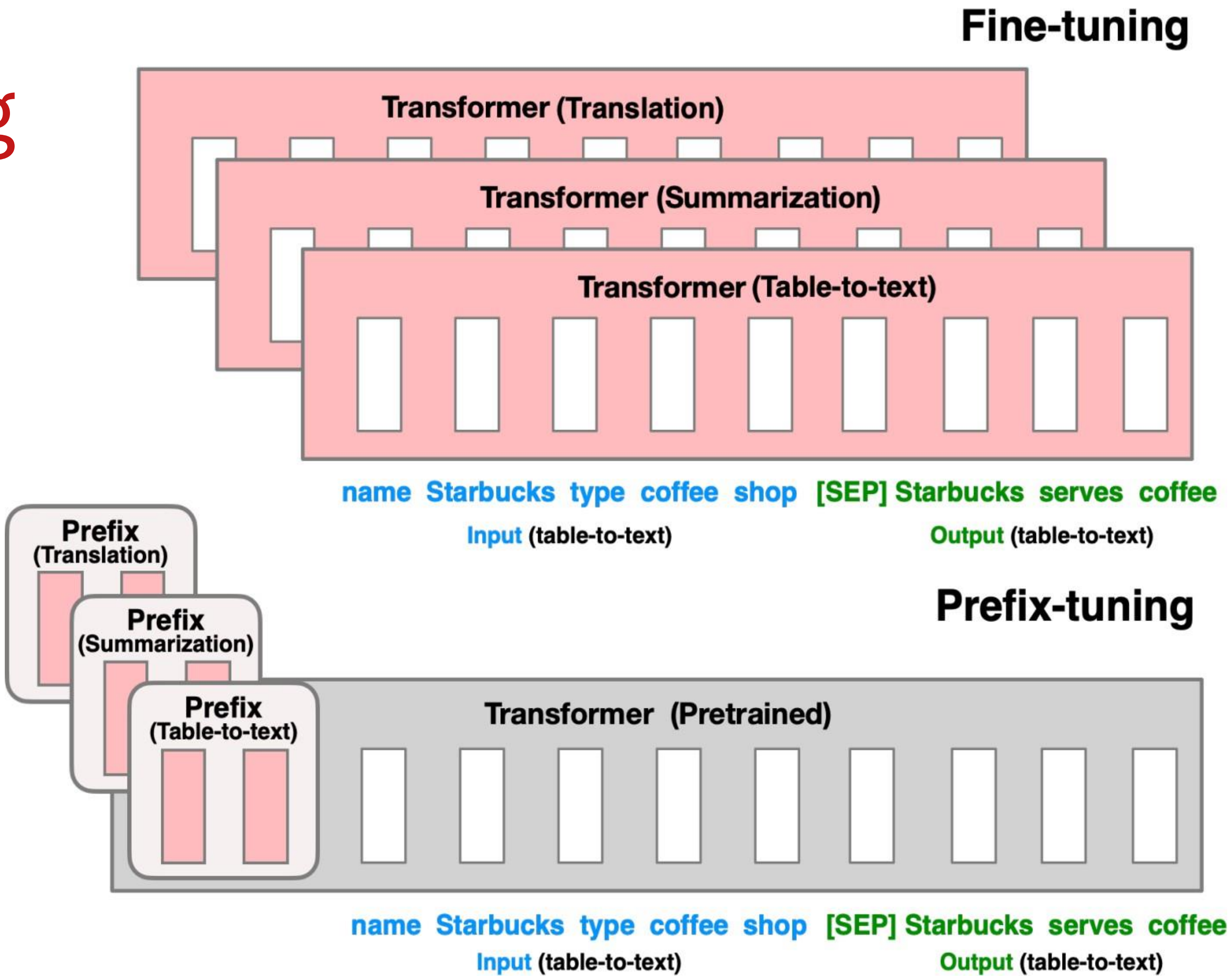


(Lester et al., 2021)

Prompt tuning

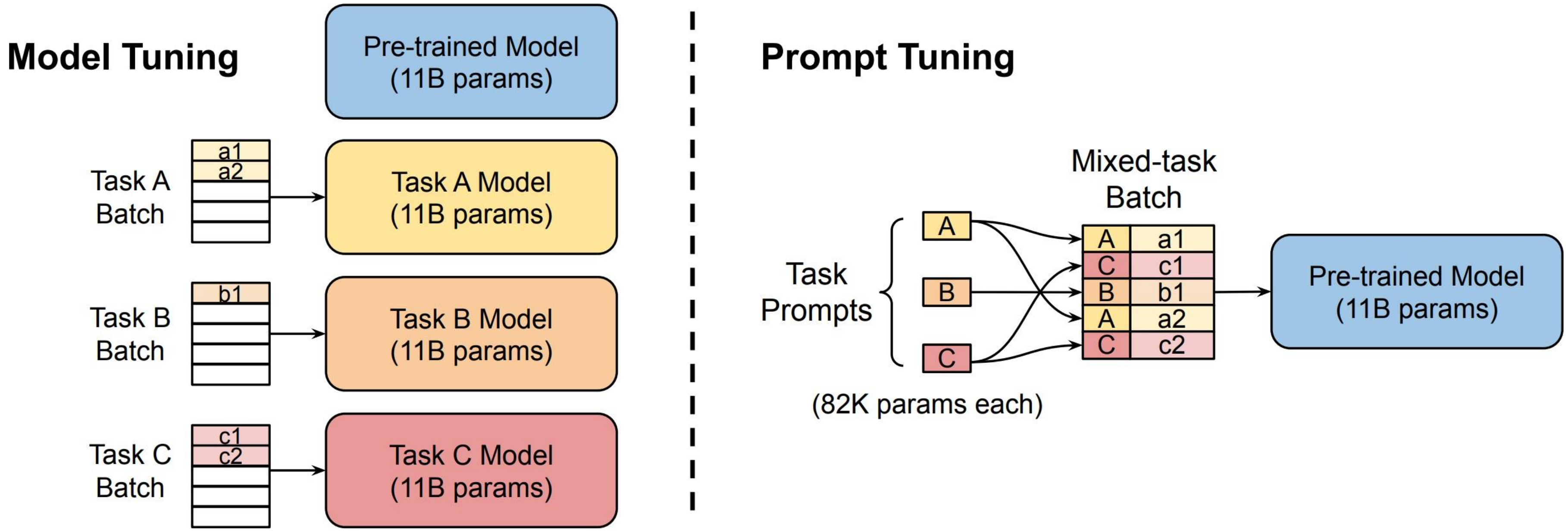


Prefix Tuning



Li & Liang, ACL 2021

Parameter-efficient Prompt Tuning



Additive

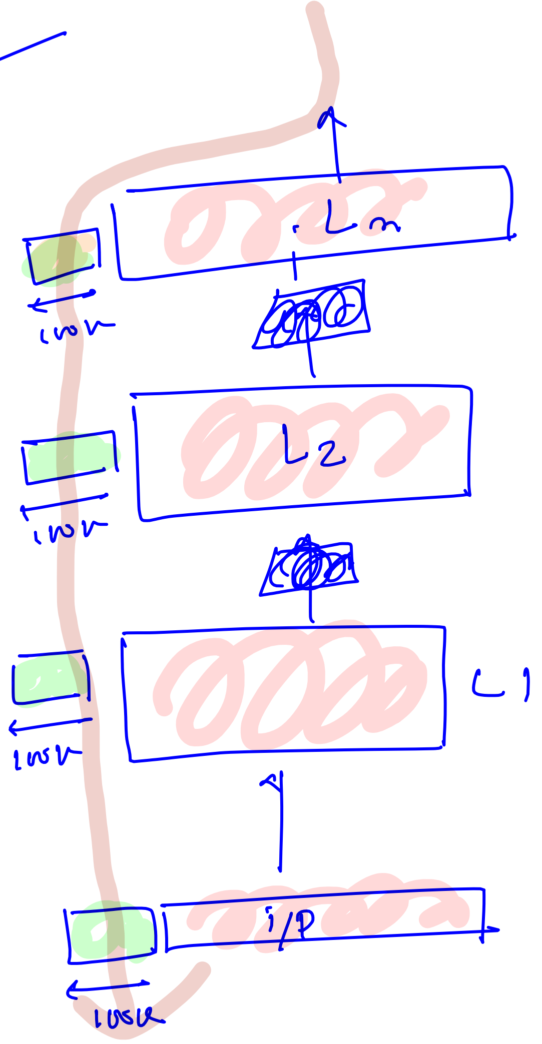
Adaptors

LORA, DoRA, Ada LORA

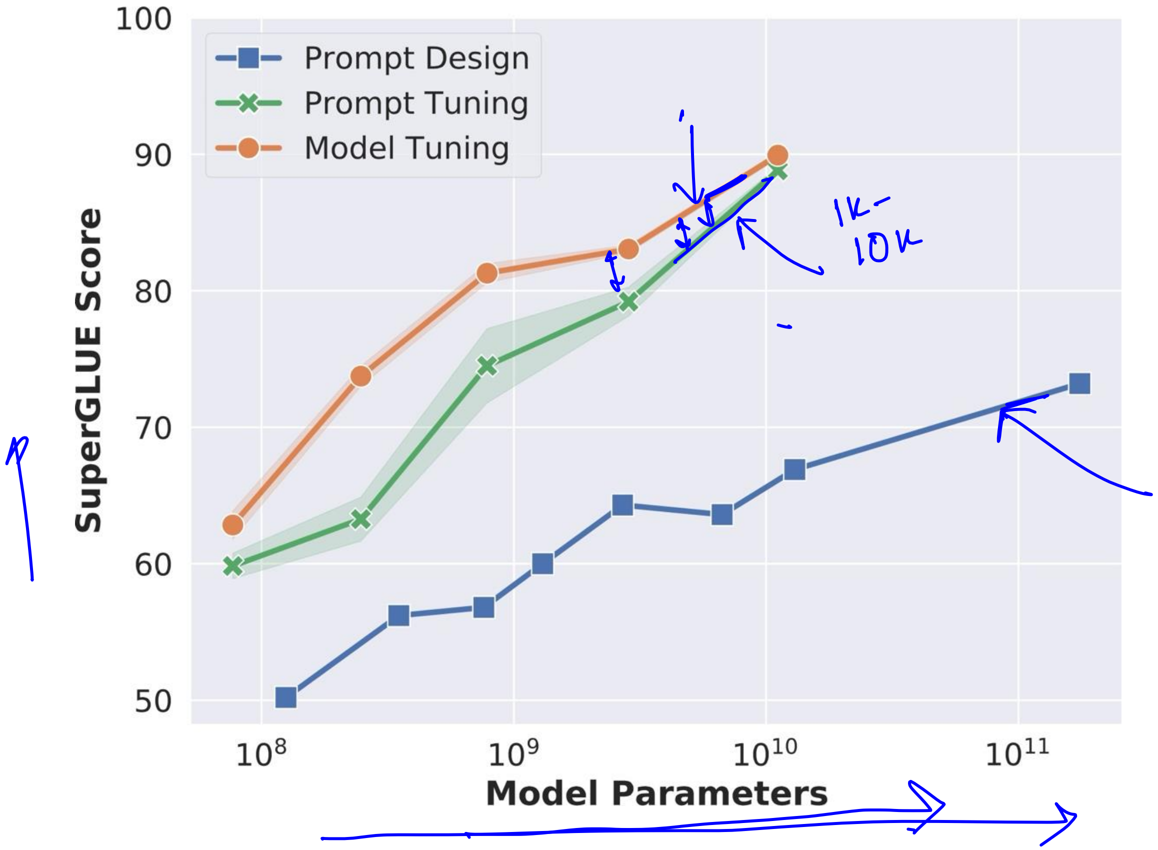
Selective

Reparameterization

Parameter-efficient Fine tuning



Prompt Tuning Becomes More Competitive With Scale



Problems With Soft Prompts

- Requires separate training
- Not possible to get soft prompts for all possible tasks and inputs
- Not user-friendly
  - How will non-expert users get soft prompts for new tasks/inputs while interacting with the LMs?

Hard prompts, thus, continue to be the default choice for interacting/utilizing LLMs.

Advanced Prompting

Prompting vs CoT

Ex, Am, Ex2: Am, Ex, Am.. Q

" Let's think step-by-step "

Q

Standard Prompting

2022

Chain-of-Thought Prompting (CoT)

Model Input

Q: Mohit has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and brought 6 more how many apples do they have?

Model Output

A: The answer is 27.

Model Input

Q: Mohit has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Mohit started with 5 balls. 2 cans of 3 tennis balls 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and brought 6 more how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9.

Prompting vs CoT

Standard Prompting

Model Input

Q: Mohit has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and brought 6 more how many apples do they have?

Model Output

A: The answer is 27.

Chain-of-Thought Prompting

Model Input

Q: Mohit has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

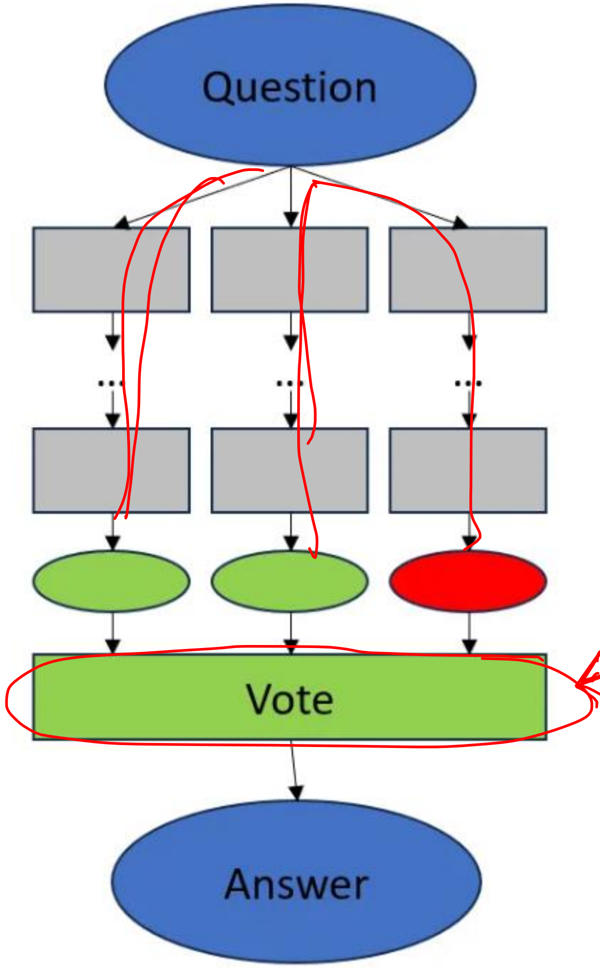
A: Mohit started with 5 balls. 2 cans of 3 tennis balls 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and brought 6 more how many apples do they have?

Model Output

A: The cafeteria had 23 appl… iginally. They used20 to … ke lunch. S… y had 23 - 20 = 3. They bou… 6 mor… les, so they have 3 + 6 = 9. The an…

CoT with Self Consistency



Procedure

1. Add „think step-by-step“ to your original question (we’ll call this augmented question the *question* in the following).
2. Ask the question repeatedly (n times) and collect the answers.
3. Decide for a voting technique and decide which of the collected answers is picked as the final answer.

https://medium.com/@johannes.koeppern/self-consistency-with-chain-of-thought-cot-sc-2f7a1ea9f941

LLM-as-a-Judge

Tree-of-Thought (ToT)

- **Key components:**
  - **Branching:** Generates multiple thought paths for each step
  - **Scoring:** Evaluates quality of each thought/path
  - **Backtracking:** Returns to previous points if a path is unproductive

4, 9, 10, 13

10-4=6

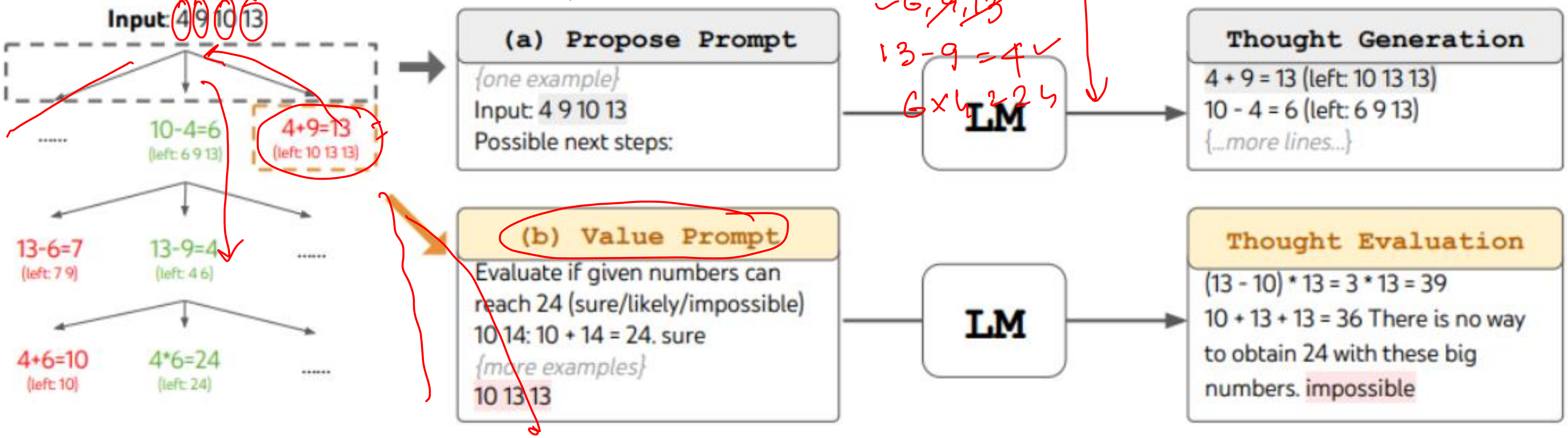
24

4, 9, 10, 13

-6, 9, 13

13-9=4

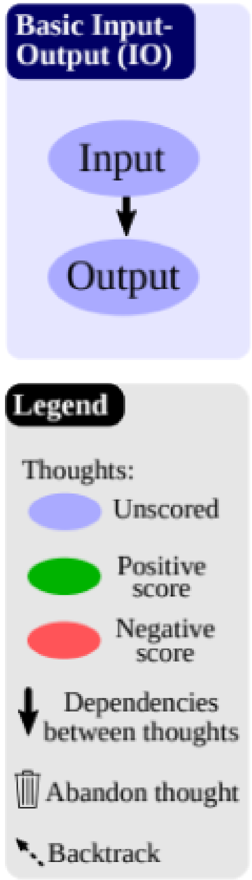
6×4=24

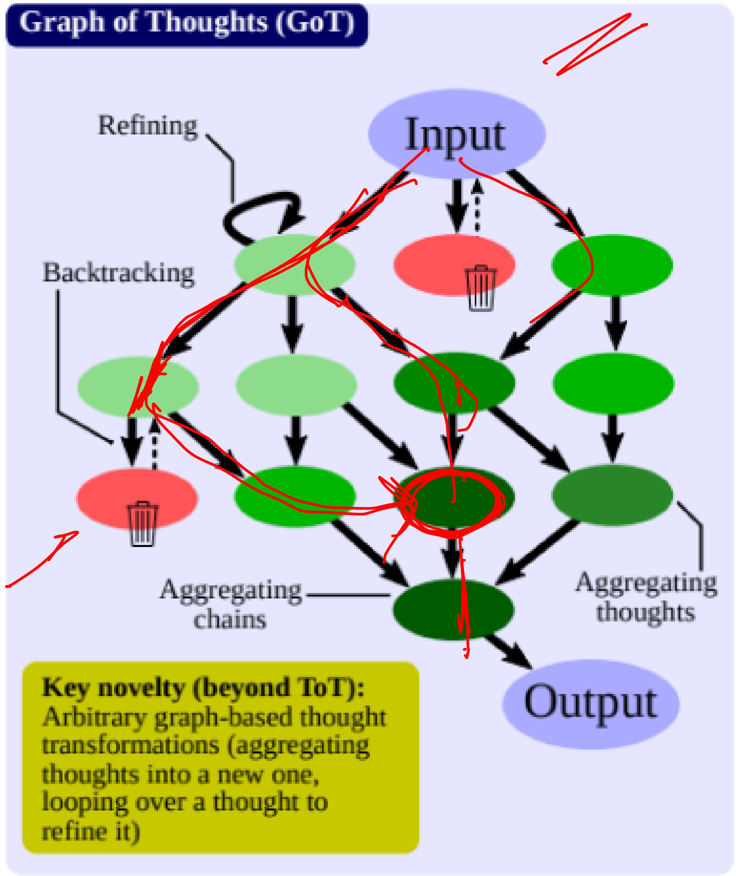


https://wandb.ai/sauravmaheshkar/prompting-techniques/reports/Chain-of-thought-tree-of-thought-and-graph-of-thought-Prompting-techniques-explained---Vmlldzo4MzQwNjMx

Graph-of-Thought (GoT)

- **Refining:** Modifies existing thoughts by adding loops in the graph
- **Aggregating:** Combines multiple thoughts into new ones by creating vertices with multiple incoming edges

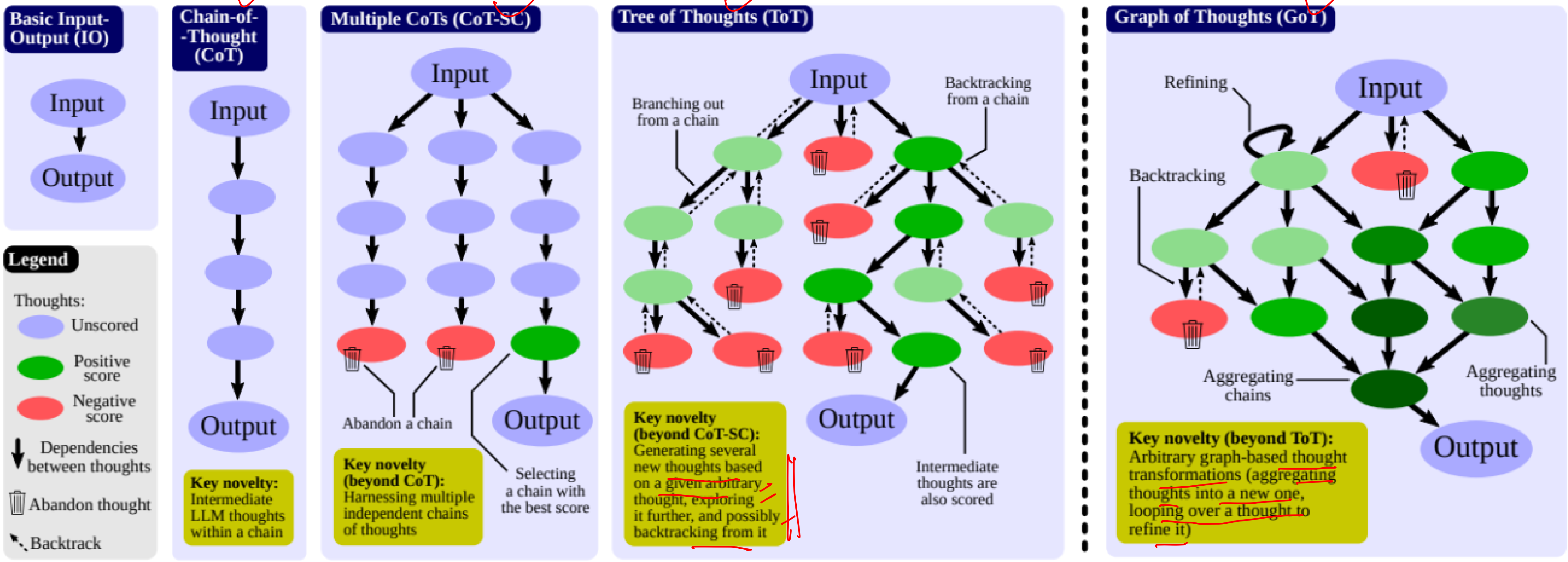




https://wandb.ai/sauravmaheshkar/prompting-techniques/reports/Chain-of-thought-tree-of-thought-and-graph-of-thought-Prompting-techniques-explained---Vmlldzo4MzQwNjMx

Graph-of-Thought (GoT)

- **Refining:** Modifies existing thoughts by adding loops in the graph
- **Aggregating:** Combines multiple thoughts into new ones by creating vertices with multiple incoming edges



https://wandb.ai/sauravmaheshkar/prompting-techniques/reports/Chain-of-thought-tree-of-thought-and-graph-of-thought-Prompting-techniques-explained---Vmlldzo4MzQwNjMx